PRACTICAL ASSESSMENT REPORT

Setting Up the n8n Advanced AI Agent Environment

Assessment date: 10 November 2026 **Platform:** n8n **LLM provider used:** Groq

Result: A functional chat-triggered AI Agent workflow was configured and successfully executed. The execution debugger confirmed successful operation of the trigger, AI Agent, and Groq Chat Model.

1. Objective and scope

Build and validate an n8n AI Agent workflow using secure LLM credentials, a configured chat model, a controlled response-length setting, and execution monitoring. The assessment test prompt was: **Hello, please introduce yourself in exactly one sentence.**

2. Implementation summary

Component	Implemented configuration
Workflow	When Chat Message Received -> AI Agent; Groq Chat Model attached to the AI Agent Chat Model port
Provider and credential	Groq; credential selected in n8n as "Groq account 7"
Model	openai/gpt-oss-20b (served via Groq)
Sampling temperature	0.0
Maximum number of tokens	500
Test result	Successful execution; one-sentence response
Model usage displayed	219 tokens in the execution log

3. Assessment compliance note

Provider deviation: The original assessment explicitly requests OpenAI or Anthropic API credentials. This implementation uses Groq because the OpenAI API account had no remaining credits. The Groq workflow demonstrates the same technical workflow concepts, but **assessor approval is required** to treat this as satisfying the provider-specific requirement.

4. Workflow construction

1. Opened the n8n workspace and created the workflow "Topic 1 - Advanced AI Agent".
2. Added the When Chat Message Received trigger and AI Agent node.
3. Connected the trigger to the AI Agent main input.
4. Attached Groq Chat Model to the AI Agent Chat Model connector.
5. Selected Groq credentials and configured the model, token limit, and temperature.

5. Model configuration evidence

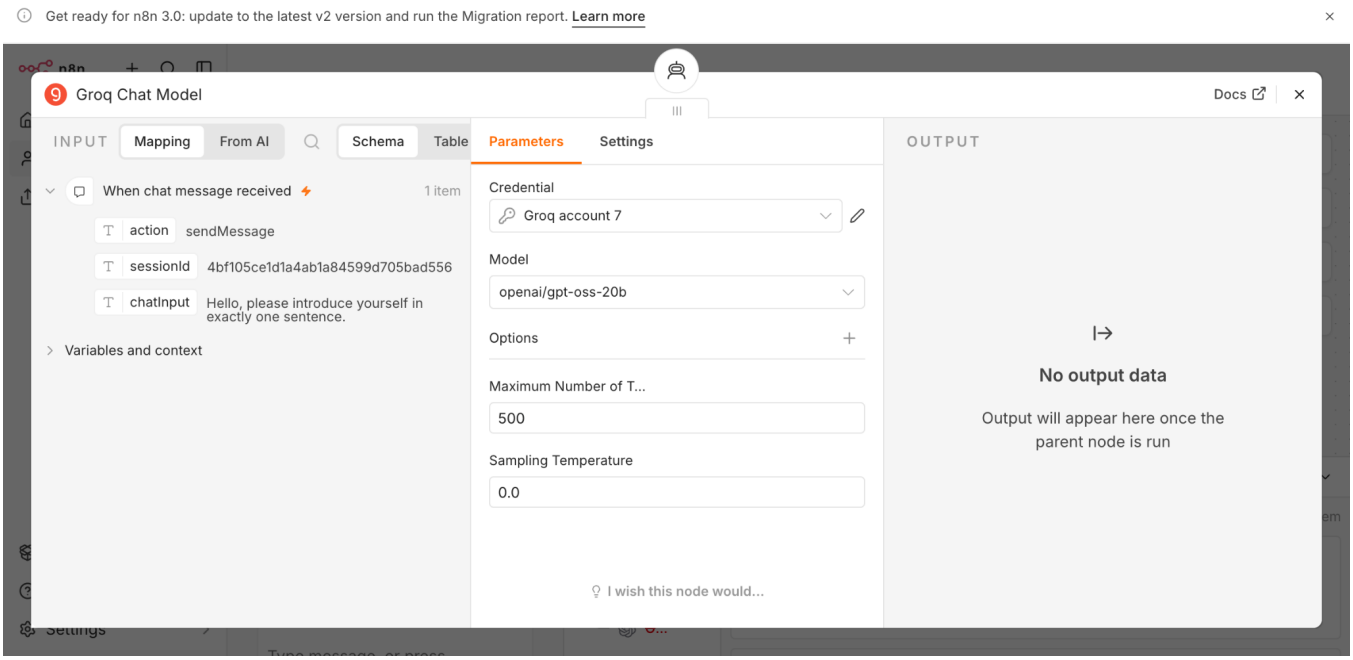


Figure 1. Groq Chat Model: selected credential, openai/gpt-oss-20b, maximum tokens 500, and sampling temperature 0.0.

6. Execution test and observed response

Input: Hello, please introduce yourself in exactly one sentence.

Output: I am ChatGPT, an AI language model developed by OpenAI, designed to assist with information, conversation, and creative tasks.

The observed output contains one sentence. The execution interface reported "Workflow executed successfully" and displayed 219 tokens for the Groq Chat Model.

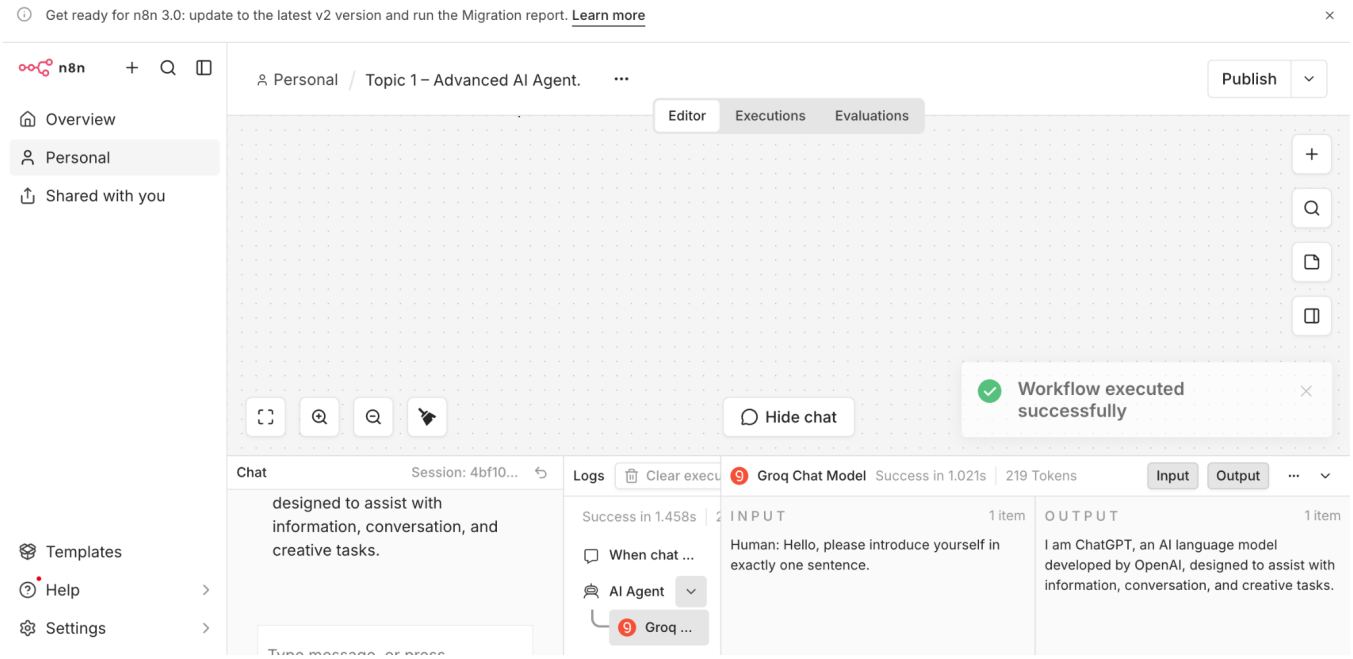


Figure 2. Successful chat execution and Groq model input/output in n8n.

7. Debugger validation evidence

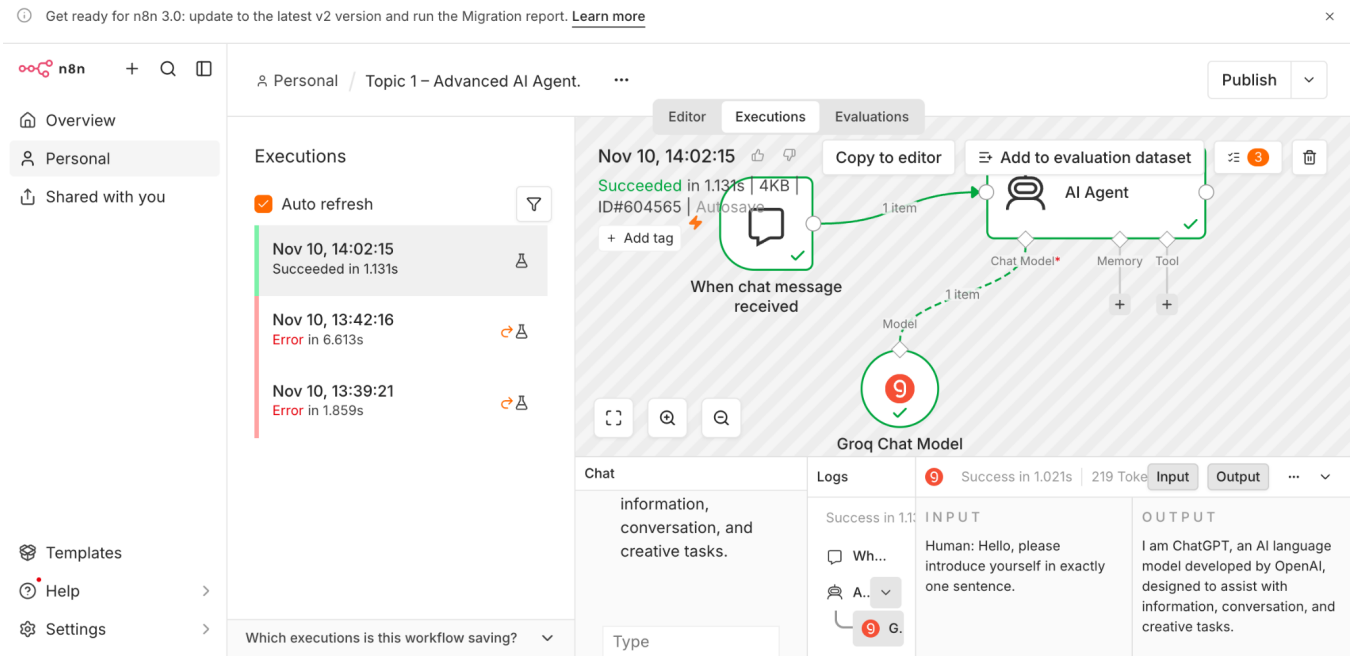


Figure 3. n8n Executions tab: latest run succeeded; all three nodes show green success indicators.

Latest execution shown: 10 November, 14:02:15 (as displayed in n8n). **Overall duration:** 1.131 seconds. **Model usage:** 219 tokens. Earlier failed attempts are retained in the execution history.

8. Troubleshooting and resolution

Observed issue	Cause / resolution
Unsupported temperature parameter	Initial OpenAI model gpt-5-mini rejected the configured temperature parameter. The model choice was reconsidered.
OpenAI API rate limit / no credits	Execution log explicitly stated that no API credits remained. Switched to Groq to continue testing without purchasing OpenAI credits.
Final validation	Groq model and secure credential were configured; workflow and debugger showed successful execution.

9. Credential security

Credentials were selected through n8n's credential management interface rather than embedding API keys in workflow logic. API keys must never appear in screenshots, documentation, exported JSON, or public GitHub repositories. Before publishing, inspect the exported JSON for embedded secrets and remove any sensitive data.

10. Execution lifecycle and token usage

Execution lifecycle: A chat message triggers the workflow; the AI Agent receives the message; the connected Groq Chat Model processes the prompt; the Agent returns a response; n8n records the execution and node-level input/output details.

Token usage: Tokens represent text units processed by a model. Input and generated output both contribute to usage. The screenshot shows a total of 219 tokens for the Groq Chat Model invocation; it does not independently establish the input/output token split.

11. Deliverables and submission status

Deliverable	Status / next action
Working workflow and test	Completed; success verified in n8n
Execution screenshots	Evidence captured in this report
Workflow JSON export	Pending - export from n8n workflow menu
Public GitHub repository	Pending - upload workflow JSON, README, documentation, and screenshots
Loom / YouTube demo	Pending - record setup, model parameters, successful run, and debugger
Assessor approval for Groq	Pending - required due to OpenAI/Anthropic provider specification

12. Suggested demonstration script

"I created a chat-triggered AI Agent workflow in n8n and connected a Groq Chat Model using credentials stored in n8n. I selected openai/gpt-oss-20b, set the temperature to zero and the maximum tokens to 500, and executed the prescribed one-sentence introduction prompt. The workflow succeeded, all three nodes showed successful execution, and the debugger displayed the model input, output, duration, and token usage. I used Groq because the OpenAI API account had no available credits, and I will confirm this substitution with the assessor."

13. Final conclusion

The practical workflow build, model configuration, execution, and debugger validation were completed successfully using Groq. Final assessment submission remains dependent on exporting the workflow, publishing the required repository and video, and obtaining approval for the alternative provider.